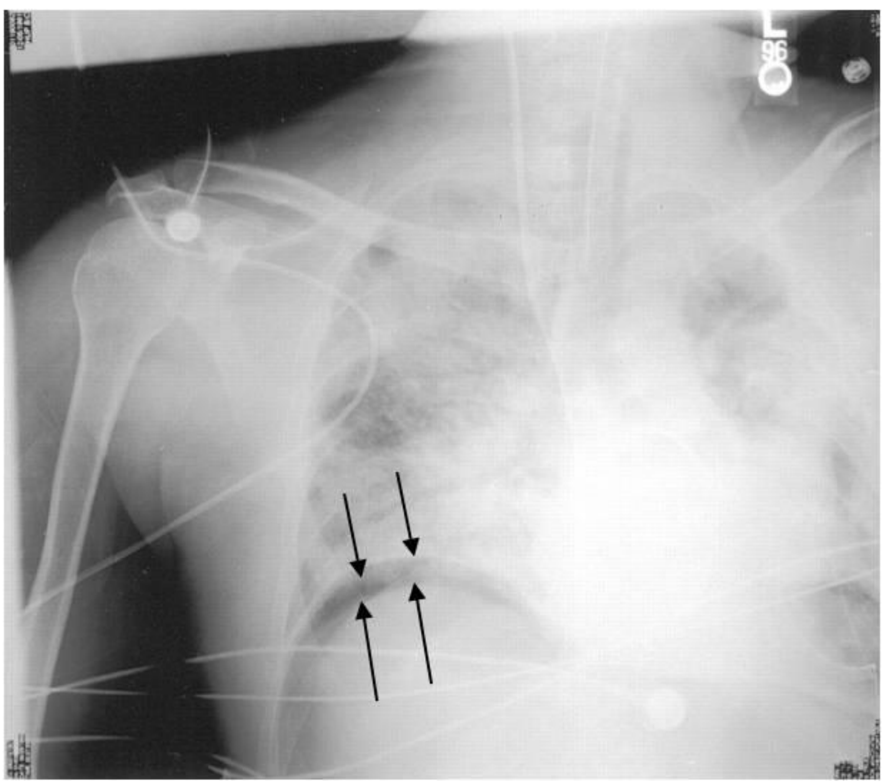
BPM2_ESOFT QUIZ 4_S2026_GD

# of Questions: 30

Question #: 1

A 44-year-old man comes to the emergency department because of a 2-hour history of a puncture wound to the abdomen. Physical examination shows the anterior wall of the stomach is punctured (leading to the release of fluid and air into the abdominal cavity). A radiograph is taken and is shown in the attached image. In which of the following areas will you most likely find the collection of air in the abdomen of this patient?

A. Hepatoorenal recess
B. Paracolic gutters
C. Subphrenic recess
D. Lesser sac
E. Rectovesical pouch



Question #: 2

A 55-year old-man comes to the physician because of a 2-month history of severe abdominal pain that worsens after a fatty meal. He is scheduled for endoscopic retrograde cholangiopancreatography, where a fiberoptic scope is passed orally to the stomach, and into the small bowel. A cannula is then inserted into the major duodenal papilla in order to visualize the biliary tree. Which of the following locations will the tip of the endoscope most likely be found in order to enter the papilla?

A. 1st part of the duodenum
B. 2nd part of the duodenum
C. 3rd part of the duodenum
D. 4th part of the duodenum
E. Duodenojejunal junction

Question #: 3

A 45-year-old man is brought to the emergency department because of a 1-hour history of hematemesis. Physical examination shows severe pallor. His abdomen is distended and there are large dilated veins on the anterior abdominal wall. An anastomosis between which of the following venous structures is most likely responsible for the appearance of the patient's abdominal wall?

A. Left gastric and azygos
B. Superior and inferior rectal
C. Liver sinusoids and inferior phrenic
D. Superficial epigastric and paraumbilical
E. Left colic and renal

Question #: 4

A 25-year-old woman is preparing for a 100-meter practice sprint. She is a sprinter who has a range of blood tests regularly performed to monitor her health, nutritional state, and adaptation to training. During one of the testing sessions her plasma glucose is 90 mg/dl (70-100 mg/dL), and her insulin/glucagon ratio is low. Which of the following best describes her metabolic state?

A. Phosphorylation and activation of glycogen synthase
B. Phosphorylation and activation of AMP-kinase by glucagon
C. Inhibition of adipocyte hormone-sensitive lipase
D. Inhibition of CPT-I due to high levels of malonyl CoA
E. Elevated hepatic fructose 2,6-bisphosphate levels
F. Low activity of mitochondrial HMG-CoA synthase

Question #: 5

A 10-month-old girl is brought to the physician for a follow-up examination. She was diagnosed with MCAD deficiency by newborn screening which was followed by genetic testing. She is advised to be fed every 6-8 hours in order to prevent fasting hypoglycemia.Which of the following hepatic enzymes is most likely lacking an allosteric activator during fasting in this patient?

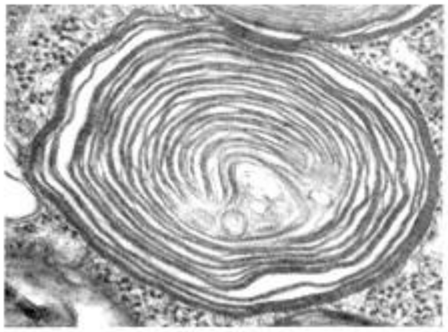
A. Glycogen phosphorylase
B. Pyruvate carboxylase
C. Phosphoenolpyruvate carboxykinase (PEPCK)
D. Alanine aminotransferase (ALT)
E. Acetyl CoA carboxylase
F. Pyruvate dehydrogenase (PDH) complex

Question #: 6

A 2-year-old boy is brought to the physician by his mother because of a 3-month history of tripping over objects and developmental delay. Physical examination shows milestone developmental delays. Examination of the skin punch biopsy specimen shows abnormal lysosomes. Which of the following compounds is most likely accumulating in this patient's lysosomes?

A. Sphingomyelin
B. Globoside
C. Ganglioside
D. Glycosaminoglycan
E. Glycogen

F. N-linked glycoprotein
G. O-linked glycoprotein



Question #: 7

A 20-year-old woman is brought to the emergency department by her friends in an unconscious state. Her respirations are 30/min and are deep and rapid. Her breath test is positive for ethanol. Acid base studies show metabolic acidosis. The high NADH/ NAD+ ratio in hepatocytes is most likely responsible for her metabolic state. Which of the following reactions is most likely favored in her liver?

A. Oxaloacetate to malate
B. Pyruvate to acetyl CoA
C. Pyruvate to oxaloacetate
D. Oxaloacetate to phosphoenolpyruvate
E. Lactate to pyruvate

Question #: 8

An 18-year-old man comes to the physician because of a 2-week history of skin lesions. He has been on a special diet containing raw egg whites for the past 3 months. Avidin present in raw eggs inhibits the absorption of biotin. Which of the following reactions is most likely impaired in this patient?

A. Conversion of homocysteine to cystathionine
B. Conversion of methylmalonyl CoA to succinyl CoA

C. Synthesis of thymidine mono phosphate
D. Conversion of phenylalanine to tyrosine
E. Synthesis of serotonin
F. Conversion of propionyl CoA to methylmalonyl CoA

Question #: 9

A 34-year-old man comes to the physician because of a 3-day history of painful skin blisters. A week earlier, he had been on a Caribbean vacation and sunbathed and drank more alcohol than usual. He developed blisters on the skin and later avoided sun exposure. His urine is red on voiding. Urinalysis is performed. Which of the following molecules would most likely be found in the highest concentrations in his urine?

A. Uroporphyrin III
B. Porphobilinogen
C. Coproporphyrin I
D. Coproporphyrin III
E. Protoporphyrin IX
F. Bilirubin
G. Homogentisic acid

Question #: 10

A 21-year-old woman comes to the physician because of a 2-month history of intermittent epigastric pain. Physical examination shows mild scleral icterus. There is no hepatosplenomegaly. Results of laboratory studies are shown in the table. Which of the following best explains the findings in this patient?

A. Decreased activity of UDP-glucuronyl transferase
B. Gall stones obstructing the common bile duct
C. Defect in the ABC transporter which secretes conjugated bilirubin
D. Increased rate of hemolysis
E. Hepatocellular damage due to viruses

	Patient	Reference range
Serum total bilirubin	3.5 mg/ dL	0.1-1.0 mg/dL
Serum direct bilirubin	3.2 mg/dL	0.1-0.3 mg/dL
Serum Aspartate aminotransferase (AST)	20 U/L	8-20 U/L
Serum Alanine aminotransferase (ALT)	20 U/L	8-20 U/L
Serum Alkaline phosphatase (ALP)	60 U/L	20-70 U/L
Urine bilirubin	Present	Absent

Question #: 11

A 10-year-old boy is brought to the physician by his parents because of a 2-day history of swelling of his feet. His mother says he may have accidentally drunk automobile antifreeze four days ago. Physical examination shows bilateral pedal edema and puffiness of his face. Laboratory studies show increased serum creatinine and increased blood urea nitrogen levels. Arterial pH is low and there is increased anion gap metabolic acidosis. Which of the following metabolites is most likely responsible for the findings in this patient?

A. Ethanol
B. Methanol
C. NAPQI
D. Acetaminophen
E. Salicylates
F. Oxalate
G. Formate
H. Acetaldehyde

Question #: 12

A 35-year-old woman is brought to the emergency department by her husband in an unconscious state. Results of laboratory studies on admission are shown in the table. Which of the following is most likely responsible for this clinical presentation?

A. Diabetic ketoacidosis
B. Hypoglycemia due to insulin overdose
C. Hyperosmolar hyperglycemic state
D. Prolonged starvation
E. Adrenergic-corticoid phase following an accident

Parameter	Patient	Reference range
Blood glucose	55	60-100 mg/dL
Ketone bodies	15	<1 mg/dL
Arterial pH	7.35	7.38-7.42
Insulin levels	5	5-25 μU/ml

Question #: 13

Laboratory and metabolic studies are performed in a 30-year-old man who has been starving for 10 days and a 30-year-old man with critical illness. Which of the following findings will most likely be observed in these individuals?

A. I
B. II
C. III
D. IV
E. V

		Starvation	Critical illness
I	Cortisol levels	High	Low
II	Metabolic rate	High	Low
III	Insulin levels	Low	Low
IV	Ketone body levels	Low	Very high
V	Blood glucose levels	Normal	High

Question #: 14

A 25-year-old man who is a downhill slalom skier is brought to the emergency department because of an accident during skiing. Corrective orthopedic surgery was performed for compound fractures to his right femur and right tibia on the day of the accident. He also has several broken ribs. There is no evidence of internal injuries. Following surgery, the patient is transferred to a post-operative ward for observation. Which of the following statements would be most accurate regarding the man's status five days after the initial accident?

A. Decreased C-reactive protein levels
B. Negative nitrogen balance
C. Decreased urinary urea
D. Decreased PGE_2 levels
E. Decreased adipocyte lipolysis

Question #: 15

A newborn is seen by the neonatologist for recurrent severe diarrhea every time he breastfeeds. A significant amount of H_2 is measured in his breath after breastfeeding. Which of the following enzyme deficiencies is the most likely diagnosis in this patient?

A. Lactase deficiency
B. Fructokinase deficiency

C. Aldolase B deficiency
D. Alpha-Amylase deficiency
E. Galactose 1-phosphate uridyl transferase deficiency
F. Galactokinase deficiency
G. Glucose 6-phosphatase deficiency

Question #: 16

A 33-year-old man is brought to the physician by his wife because of a 1-year history of anorexia and weight loss. His wife says that he has mild behavioral changes over the same period. Physical examination shows that the liver is not palpable. Eye examination shows Kayser-Fleischer (KF) rings in the cornea. Examination of a liver biopsy specimen shows cirrhosis. Which of the following best accounts for the clinical presentation?

A. Defect in biliary copper excretion
B. Defect in dietary copper absorption
C. Increase in dietary iron absorption
D. Deficiency of dietary iron uptake
E. Decreased intrinsic factor secretion from parietal cells
F. Increased incorporation of copper to ceruloplasmin
G. Homozygosity for mutation in the *HFE* gene

Question #: 17

A 38-year-old man comes to the physician because he is finding it difficult to lose weight. He feels that he is gaining more weight despite dietary modifications. His BMI is 38 kg/m^2, and waist-to-hip ratio is 1.3. Which of the following hormonal conditions is most likely contributing to this patient's uncontrolled weight gain?

A. Increased adiponectin secretion
B. Adiponectin resistance
C. Resistance to growth hormone
D. Leptin resistance
E. Elevated cortisol secretion

Question #: 18

A 35-year-old man comes to the physician because of a 3-day history of malaise and yellowing of the skin. Physical examination shows scleral icterus. Results of laboratory studies are shown in the table. Which of the following is the most likely diagnosis?

A. Obstruction of the common bile duct due to pancreatic cancer
B. Gilbert syndrome
C. Chronic alcoholic cirrhosis
D. Hemolysis due to sickling crisis
E. Acute hepatocellular damage due to viruses
F. Dubin-Johnson syndrome

Serum parameter	Patient	Reference range
Aspartate aminotransferase (AST)	800	8-20 U/L
Alanine aminotransferase (ALT)	1050	8-20 U/L
Alkaline phosphatase (ALP)	150	20-70 U/L
γ-glutamyl transferase (GGT)	70	10-30 U/L
Direct bilirubin	3.8	0-0.3 mg/dL
Total bilirubin	7.4	0.1-1.0 mg/dL
Albumin	4.2	3.5-5.5 g/dL

Question #: 19

A 2-year-old girl is brought to the physician by her parents because of a 2-month history of failure to thrive and diarrhea. Physical examination shows skin lesions in the exposed areas of the neck, hands, and feet. Urinalysis shows the presence of tryptophan. Which of the following is the best explanation for the physical examination findings in this patient?

A. Negative nitrogen balance
B. Tryptophan inhibiting serotonin synthesis
C. Tryptophan inhibiting melanin synthesis

D. Decreased NAD^+ synthesis from tryptophan
E. Accumulation of homocysteine and methylmalonate

Question #: 20

A 4-year-old girl is brought to the physician by her father because of a 2-month history of vision problems. She is taller than age and sex-matched controls. Physical examination shows lens dislocation. There are also early signs of atherosclerosis. Administration of pyridoxine results in improvement. Which of the following processes is most likely affected in this patient?

A. Catabolism of the branched chain amino acids
B. Beta oxidation of fatty acids
C. Conversion of homocysteine to cysteine
D. Synthesis of heme from glycine and succinyl CoA
E. Liver glycogenolysis

Question #: 21

A 4-month-old boy is brought to the physician by his parents because of failure to thrive since birth. He is short in stature compared to age and sex-matched controls. Physical examination shows coarse facial features, macrocephaly and hepatosplenomegaly. His joint mobility is severely restricted and there is corneal clouding. Examination of skin biopsy specimens show the presence of cytosolic inclusion bodies in the fibroblasts. Which of the following is most likely responsible for the clinical presentation in this patient?

A. Isolated defect in the degradation of gangliosides
B. Isolated deficiency of sphingomyelinase
C. Defect in the synthesis of glycosaminoglycans
D. Defect in sorting of lysosomal enzymes
E. Defect in the synthesis of sphingoglycolipids

Question #: 22

Fifty women aged group of 35 to 40 years are recruited as volunteers for the study of digestion and absorption. They are given a meal containing 40% fat. Which of these molecules is best suited for enhancing digestion and subsequent absorption of this macronutrient by the intestinal mucosal cells?

A. Deoxycholic acid
B. Chenodeoxycholic acid
C. Cholic acid
D. Taurocholic acid
E. Lithocholic acid
F. Bilirubin diglucuronide

Question #: 23

A 13-year-old girl is brought to the physician by her parents because of a 2-day history of increased frequency of urination, increased thirst and weight loss. Four days ago, she was diagnosed with a lung infection and was started on antibiotics. She responded poorly to the antibiotics and the infection did not improve. Physical examination shows dry mucus membranes. Urine dipstick tests are positive for glucose and ketone bodies. Which of the following is most likely an additional laboratory finding in this patient?

A. Very low circulating insulin levels
B. Decreased plasma glucose levels
C. Normal HbA1c levels
D. Decreased glucose output by liver
E. Increased circulating C-peptide levels
F. Increased arterial pH
G. Increased serum bicarbonate levels

Question #: 24

A 39-year-old man is brought to the emergency department because of a 4-hour history of severe abdominal pain. He has consumed excessive amounts of alcohol over the past 2 days. Physical examination shows acute tenderness to his epigastric region. Laboratory studies show elevated levels of pancreatic enzymes suggestive of acute pancreatitis. This condition is characterized by autolysis of the secretory units of the parenchyma of this gland as a result of the activity of a stimulatory hormone. Which of the following cells most likely secretes the stimulatory hormone?

A. Parietal
B. Paneth
C. Chief
D. Enteroendocrine
E. Goblet
F. Mucous

Question #: 25

A 25-year-old man comes to the physician because of a 1-month history of intermittent abdominal cramps and bloating. Physical exam shows no abnormalities. History shows the symptoms typically appear following a meal rich in carbohydrates. Genetic testing shows a mutation in a gene for an enzyme that breaks down oligosaccharides to monosaccharides for absorption. Which of the following best characterizes the localization of this enzyme?

A. Digestive organelles of hepatocytes
B. Cytoplasm of enteroendocrine cells
C. Microvilli of enterocytes
D. Glands of the intestinal submucosa
E. Lumen of the stomach

Question #: 26

A 43-year-old man comes to the physician because of a 2-week history of diarrhea and abdominal pain. He has a 3-year history of heavy alcohol consumption. Physical examination shows tenderness of the epigastric region. Laboratory studies of the blood lead to the diagnosis of chronic pancreatitis, a condition characterized by the slow autolytic destruction of the parenchyma by its own enzymes. Which hormone regulates bicarbonate secretion from this gland?

A. Gastrin

B. Secretin
C. Cholecystokinin
D. Motilin
E. Glucagon
F. Ghrelin

Question #: 27

A 58-year-old man comes to the physician because of a 2-month history of heart burn and belching. Laboratory studies show that his gastric acid output exceeds normal levels. Ultrasonography of the abdomen shows a 2.3-cm mass in the pancreas. He is suspected of having a tumor in the pancreas. This tumor secretes a large amount of gut hormone that is responsible for increased gastric acid secretion in the stomach. Which of the following gastro-intestinal cell types is most likely the source of this hormone?

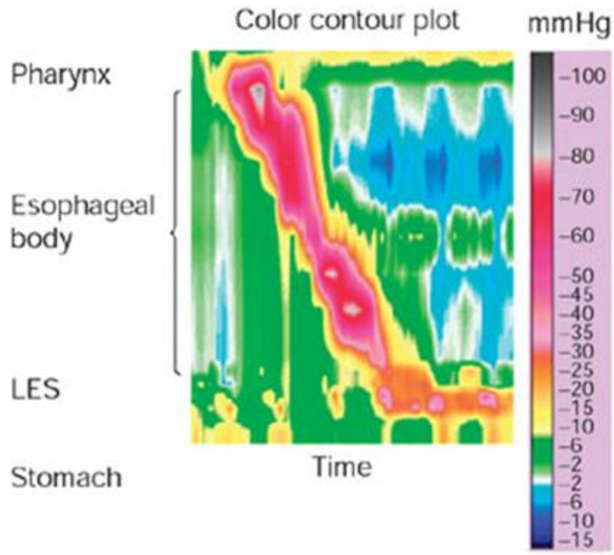
A. S cells
B. G cells
C. Chief cells
D. D cells
E. Enteroendocrine cells

Question #: 28

A 48-year-old man comes to the physician because of a 3-week history of diffuse chest pain and difficulty swallowing. His vital signs are within normal limits. Physical examination shows no abnormalities. The result of high-resolution esophageal manometry is shown in the attached color contour plot. Which of the following interpretations of this manometry result is most accurate?

A. The upper esophageal sphincter does not open
B. Primary peristalsis is absent in the esophageal body
C. The lower esophageal sphincter does not open
D. There is functional peristalsis in the upper and lower esophagus
E. Secondary peristalsis is absent in the esophageal body

[redacted]



Question #: 29

A 32-year-old man comes to the physician because of a 6-month history of recurrent episodes of burning sensation in the throat. He says that these episodes usually wake him up at night. His BMI is 34 kg/m^2 (Normal: 19-25 kg/m^2). Physical examination shows no abnormalities. Endoscopy of the upper GIT shows peptic ulcers in the proximal duodenum. He is advised to take a drug that inhibits H^+/K^+ ATPase in the stomach for 4 weeks. Which of the following effects is most likely expected in this patient after a meal?

A. Failure to increase pH of the gastric venous blood
B. Increased carbonic anhydrase activity in the stomach
C. Decreased Na^+ in gastric juice
D. Increased H^+ in gastric juice
E. Gastric juice becomes hypotonic

Question #: 30

A 33-year-old man is brought to the emergency department by his friend because of a 1-hour history of altered consciousness. His friend says he was participating in a hunger strike for the past 3 days and did not eat or drink anything. His pulse is 104/min and blood pressure is 104/62 mm Hg. Physical examination shows dry mucous membranes, sunken eyes and altered mental state. Abdominal auscultation shows occasional bowel sounds. These sounds are produced by strong propulsive contractions that pass from stomach through small intestine during fasting. Which of the following humoral substances most likely mediates this intestinal motility in the patient?

A. Acetylcholine
B. Motilin
C. Nitric Oxide
D. Vasoactive Intestinal polypeptide
E. Serotonin